



Customizing the morphology and microstructure of single-crystalline Ni-rich layered cathode materials for all-solid-state batteries

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ABSTRACT

A single-crystalline Ni-rich cathode material is synthesized for high-performance all-solid-state batteries (ASSBs). We customize the morphology and microstructure of the material by optimizing the synthesis process (particle size control and post-heat treatment) and demonstrate the advantages of single-crystalline cathode materials for practical use in ASSBs. The particle density and particle hardness of single-crystalline cathode materials with a particle size of 5 μm are considerably higher than those of the conventional polycrystalline cathode materials that are composed of spherical aggregates of submicron-scale primary particles. These structural features of the single-crystalline cathode materials improve the long-term cycling performance and rate capability of an ASSB configured with an argyrodite- $\text{Li}_6\text{PS}_5\text{Cl}$ solid electrolyte. This is mainly attributed to the intimate interfacial solid-solid contacts and enhanced diffusion kinetics induced by the monolithic morphology of the single-crystalline cathode materials in the electrode. Moreover, undesirable microcrack formation and particle fracture in cathode materials are effectively suppressed during cycling. The feasibility of ASSBs configured with single-crystalline Ni-rich cathode materials is examined in detail.

1. Introduction

The development of advanced lithium-ion batteries (LIBs) has become crucial owing to the global growth in electric vehicle (EV) markets [1]. Although LIBs are the most widely used power sources for EVs, it is challenging to achieve high driving mileage and ensure safety. In particular, safety failures frequently occur during the abnormal operation of LIBs, such as overcharging, overheating, and external short circuits [2–5]. Thus, considerable attention has been paid to the development of highly reliable energy storage systems that meet the rigorous industrial standards for EV applications [6].

In recent years, all-solid-state batteries (ASSBs), which are configured with nonflammable solid electrolytes replacing conventional liquid electrolytes, have been highlighted as future energy solutions because of their high energy density and excellent safety [5,7,8]. High-performance ASSBs require highly reliable solid electrolytes that offer high ionic conductivity comparable to that of conventional liquid electrolytes ($\sim 10^{-2} \text{ S cm}^{-1}$) [9–12]. Moreover, Li^+ transport pathways must be

designed at the electrode level. Unlike conventional LIBs, Li^+ transport occurs over the solid bulk and across the solid/solid interface in ASSBs. Therefore, the interfacial engineering of the solid-solid interfaces between solid electrolytes and cathode materials is a prerequisite for ensuring sufficient Li^+ transport pathways in an electrode [13–15]. Hence, it is essential to customize the morphology and microstructure of cathode materials for establishing robust Li^+ pathways to enhance the electrochemical performance of ASSBs.

Recently, Ni-rich cathode materials are commercially utilized in conventional LIBs owing to their high reversible capacity ($> 200 \text{ mA g}^{-1}$) [16–21]. Unfortunately, the practical use of these cathode materials in ASSBs is limited because of their relatively low density and particle cracking and loss of Li^+ transport pathways, resulting in significant performance fading in ASSBs during cycling [22–29]. Particle cracking in ASSBs must be suppressed because solid electrolytes cannot penetrate particles after particle cracking, leaving certain parts of fractured particles isolated. These electrochemically inactive parts induce significant capacity fading in ASSBs during cycling [30,31].

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In this respect, various structural modifications have been explored to enhance the mechanical strength and density of cathode materials to successfully implement them in ASSBs [26–29]. From this perspective, single-crystalline cathode materials are potential candidates for ASSBs because of their monolithic (grain-boundary-less) morphology, defect-free microstructure, and excellent particle hardness [31–33]. The unique structural features of single-crystalline cathode materials are beneficial for constructing stable and fast Li^+ transport pathways by reducing grain boundaries and voids inside particles and suppressing inevitable particle cracking during electrode fabrication and cycling.

Herein, the feasibility of single-crystalline Ni-rich cathode materials is examined in detail using an argyrodite- $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl) solid electrolyte through various structural and electrochemical investigations. Unlike conventional cathode materials, single-crystalline cathode materials are beneficial for constructing robust Li^+ transport pathways in an electrode, thereby improving the cycling performance and rate capability of ASSBs. The superior density and mechanical strength of single-crystalline cathode materials can effectively suppress the undesirable performance fading in ASSBs. We suggest that it is essential to customize the cathode materials to make them suitable for use in ASSBs, along with a comprehensive understanding of the Li^+ transport mechanism, to develop high-performance ASSBs.

2. Experimental

2.1. Material preparation

First of all, $\text{Ni}_{0.82}\text{Co}_{0.12}\text{Mn}_{0.06}(\text{OH})_2$ powders were initially prepared via a conventional co-precipitation process. After mixing with a stoichiometric amount of lithium hydroxide (LiOH), the mixture was calcined at temperatures of 800–900 °C for 24 h in a sealed tube furnace under an O_2 atmosphere. A wet-milling process and post-heat treatment were subsequently adopted to control the morphology and microstructure of single-crystalline Ni-rich cathode materials. The as-prepared cathode material (2 g) was wet milled with an isopropyl alcohol solution at 150 rpm for 20 min, and then, the milling was paused for 10 min. This process was repeated six times. The resulting powder was further dried at 120 °C in a vacuum oven until the solvent was fully evaporated. Finally, a post-heat treatment was applied for structural recovery at temperatures of 600–700 °C for 6 h under an O_2 atmosphere.

2.2. Structural characterizations

The morphologies and microstructures of as-prepared cathode materials were characterized by field-emission scanning electron microscopy (FESEM, JSM-7000F, JEOL) coupled with energy-dispersive X-ray spectroscopy (EDS) and transmission electron microscopy (TEM, ARM-200F, JEOL), respectively. For observation of the cross-sectional SEM images, the electrodes were polished using a cooling cross-sectional polisher (IB-19520CCP, JEOL) at –50 °C and then observed by using SEM (JSM-IT200, JEOL) in a dry room at room temperature.

Powder X-ray diffraction (XRD, Empyrean, PANalytical) was employed to confirm the phases of the cathode materials. A micro-indenter (MCT-W500, Shimadzu) and particle size analyzer (PSA, S3500 series, MICROTRAC) were used to examine the particle hardness and particle size distribution of the cathode materials, respectively. Moreover, the surface area and tap density were measured using a porosity analyzer (TriStar II 3020, Micromeritics) and tap density analyzer (Quantachrome, FL33426, BOYNTON BEACH), respectively. An inductively coupled plasma mass spectroscopy (ICP-MS, Aurora M90, Bruker) was used for compositional verifications.

2.3. Electrochemical measurements for LIB

A slurry composed of the cathode materials (90 wt%), a conducting agent (5 wt%, Super P), and a polyvinylidene fluoride binder (5 wt%) in

an N-Methyl-2-pyrrolidone solution was prepared for electrode fabrication and then coated on a current collector (15 μm thick Al foil). The electrodes were roll pressed at 200 kg cm^{-2} after drying at 120 °C for 12 h in a vacuum oven. The loading amount of the cathode material was 17.0 mg in the electrodes with an area of 1.131 cm^2 . To evaluate the electrochemical performance, CR2032 coin-type half cells were assembled with a Li metal foil and a polyethylene separator in a dry room. As an electrolyte, 1 M LiPF_6 dissolved in ethylene carbonate and ethyl methyl carbonate (1:2 v/v) was used for cell assembly. Electrochemical tests were performed at various current densities of 0.1–5.0C (1C = 200 mA g^{-1}) with different cut-off voltages (4.3 and 4.5 V vs. Li/Li^+).

2.4. Electrochemical measurements for ASSB

Composite electrodes were prepared by mixing the cathode materials (60 wt%), a conducting agent (5 wt%, Super P), and a solid electrolyte (35 wt%, $\text{Li}_6\text{PS}_5\text{Cl}$). The composite (10 mg) and Li–In foil were placed on either side of a solid electrolyte pellet (100 mg, 10 mm in diameter). The cathode/electrolyte stack was pelletized by applying an external pressure of 437 MPa. Then, the Li – In foil was placed on the other side before sealing with a torque of 11 N m. The ASSB was assembled in the stacking sequence of Al foil/cathode/solid electrolyte/Li – In/Cu foil. The electrochemical performance of the ASSB was evaluated using a galvanostatic cycler (TOYO TOSCAT-3100) after applying external pressure. The cell assembly and electrochemical tests were performed using a hermetically sealed cell in a dry room. The electrochemical tests were performed at current densities of 0.05–5.0C (1C = 200 mA g^{-1}) with different cut-off voltages (3.7 and 3.9 V vs. In-Li/Li^+).

2.5. Electrochemical analysis

Galvanostatic intermittent titration technique (GITT) measurements were conducted for 10 min at a constant current of 0.1C and then relaxed for 1 h using a galvanostatic cycler (WonA Tech. WBCS3000L). Nyquist plots were collected using an impedance analyzer (VSP-300, Bio-Logics) with an amplitude of 15 mV in a frequency range from 7 MHz to 10 mHz. All the measurements were performed at 30 °C.

3. Results and discussion

On the basis of the reaction mechanism of ASSBs, the morphology and microstructure of cathode materials must be customized for boosting the electrochemical performance at the interfaces between solid electrolytes and cathode materials [34–37]. To develop a high-performance ASSB, the fundamental issues regarding the interfacial reactions that occur at the solid–solid interfaces must be resolved. Thus, we prepared polycrystalline $\text{LiNi}_{0.82}\text{Co}_{0.12}\text{Mn}_{0.06}\text{O}_2$ (PC-NCM) and single-crystalline NCM (SC-NCM) with a high Ni concentration of 82 at% to elucidate the importance of the structural modification of cathode materials in ASSBs.

Fig. 1 compares the schematics of composite cathodes containing cathode materials with different morphologies and microstructures for ASSBs. PC-NCM and SC-NCM particles are embedded in an LPSCl solid electrolyte through the formation of solid–solid interfaces. Unlike the PC-NCM particles, which are composed of multiple primary particles (Fig. 1a), the SC-NCM particles have a typical monolithic morphology with a relatively smooth surface (Fig. 1b). The PC-NCM particles contain numerous grain boundaries and internal voids between primary particles, which act as bottlenecks for concurrent Li^+ and electron transport and increase the resistance for chemical (ambipolar) diffusion during the operation of ASSBs [38–40]. In contrast, the SC-NCM particles are characterized by the absence of grain boundaries and internal voids, along with enhanced interfacial solid–solid contacts. This facilitates the effective transport of Li^+ and electrons through the particles and across the SC-NCM/LPSCl interfaces. Such morphological and microstructural differences in the cathode materials directly affect the electrochemical

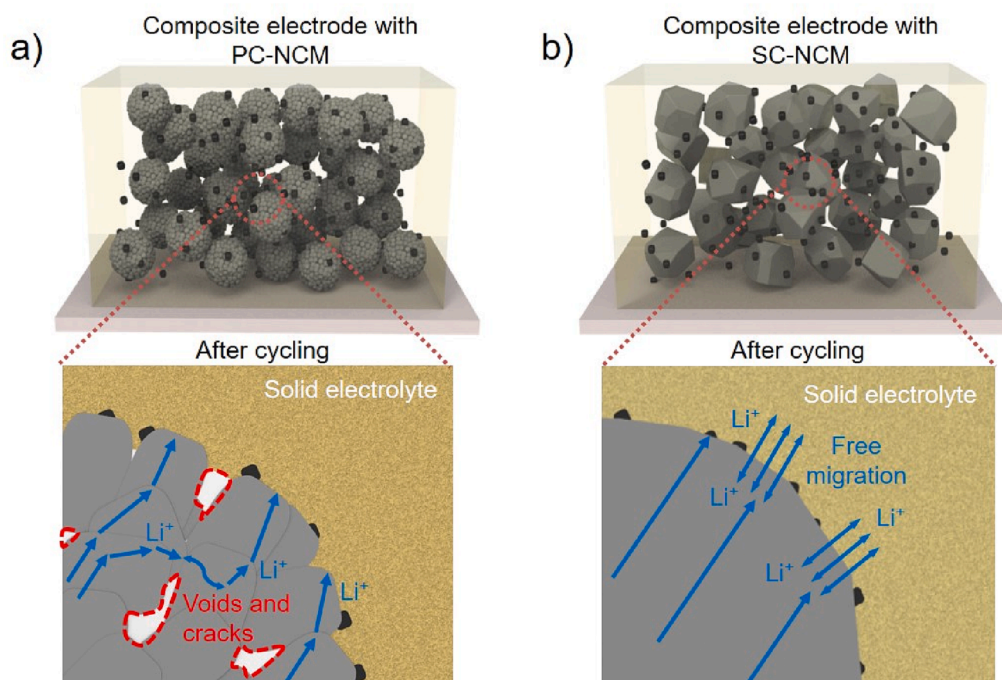


Fig. 1. Schematic of Li^+ transport in composite electrodes with different types of NCMs for ASSBs: (a) PC-NCM and (b) SC-NCM particles embedded in LPSCI solid electrolytes.

performance of ASSBs.

The morphologies of the $\text{Ni}_{0.82}\text{Co}_{0.12}\text{Mn}_{0.06}(\text{OH})_2$ precursor and synthesized NCM particles are shown in Fig. 2 and Fig. S1. The spherical $\text{Ni}_{0.82}\text{Co}_{0.12}\text{Mn}_{0.06}(\text{OH})_2$ particles with a particle size of $\sim 5 \mu\text{m}$ were successfully prepared through the coprecipitation process after the optimization of process variables (Fig. 2a). As confirmed by elemental mapping results using EDS, transition metals (TMs) were uniformly distributed in the particles (Fig. S2), representing a high fraction of Ni ($>82 \text{ at}\%$). Both PC-NCM and SC-NCM cathode materials were synthesized with stoichiometric amounts of LiOH at different temperatures. After heating at 800°C , spherical PC-NCM particles composed of primary particles ($\sim 600 \text{ nm}$) were obtained with a particle size of $D_{50} = 4.9 \mu\text{m}$ (Fig. 2b).

Additional processes were required to synthesize the SC-NCM cathode materials. The synthesis temperature was increased to 900°C to obtain a monolithic morphology of SC-NCM particles through the growth of primary particles. However, as-prepared SC-NCM particles suffered from an undesirable aggregation of monolithic particles with irregular particle shapes and a particle size of $D_{50} = 13.0 \mu\text{m}$ (Fig. 2c). The large size of as-prepared SC-NCM particles is kinetically unfavorable for Li^+ transport because of its long mean free path. An additional wet-milling process was employed to mitigate particle aggregation during the synthesis of SC-NCM particles to further control their morphology. The size of wet-milled SC-NCM particles was considerably smaller ($\sim 4.5 \mu\text{m}$), as presented in Fig. 2d. Even though the wet-milling process leads to effective particle separation, it inevitably damages the structure of SC-NCM particles because of mechanical forces.

In order to recover the structural damage to surface of particles, a post-heat treatment was applied to the wet-milled SC-NCM particles. It was crucial to determine the optimal post-heat treatment temperature to prevent particle agglomeration. In this respect, the wet-milled SC-NCM particles were carefully heated in the temperature range of 600 to 700°C for 6 h under an O_2 atmosphere. The particle size distribution was analyzed using PSA, while the morphology of the particles was observed using FESEM (Fig. S3). From the results, we found that the particle size of SC-NCM particles was maintained up to 675°C with a value of $D_{50} = 5 \mu\text{m}$. However, the D_{50} value increased to $7 \mu\text{m}$ at 700°C , indicating

particle agglomeration. Furthermore, the FESEM images confirmed that the surface damage was effectively recovered by the post-heat treatment of the wet-milled SC-NCM particles.

To determine the optimal temperature of the post-heat treatment, the electrochemical performance was evaluated in LIB assembled with a conventional liquid electrolyte. Galvanostatic voltage profiles were measured from 3.0 to 4.3 V vs. Li/Li^+ at various current densities ranging from 0.1 C to 5.0 C (Fig. S4). The SC-NCM heated at 600°C exhibited lower rate capability compared to the as-prepared SC-NCM, suggesting that this temperature was insufficient to effectively recover the surface damage. As the temperature increased to 675°C , the SC-NCM showed enhanced rate capability compared to the as-prepared SC-NCM. In the case of SC-NCM heated at 700°C , particle agglomeration was evident. Therefore, it can be inferred that an optimum temperature of 675°C offers morphologically well-defined SC-NCM particles with a particle size of $D_{50} = 5 \mu\text{m}$ without any structural damage (Fig. 2e and 2f).

The surface areas of SC-NCM and PC-NCM particles were estimated from N_2 isotherms as 0.23 and $0.28 \text{ m}^2 \text{ g}^{-1}$, respectively (Fig. S5a). The tap density of SC-NCM particles (2.10 g cc^{-1}) was higher than that of PC-NCM particles (2.02 g cc^{-1}) (Fig. S5b). This showed that SC-NCM particles are considerably denser than conventional PC-NCM particles. The corresponding EDS elemental mapping results confirmed the chemical composition and homogeneity of the TMs in SC-NCM particles (Fig. S2). The chemical composition of SC-NCM particles was determined using ICP-MS, which revealed a TM ratio of $\text{Ni}:\text{Co}:\text{Mn} = 82:12:6$.

Fig. 3a compares the powder XRD patterns of PC-NCM and SC-NCM particles. All reflections were indexed to the typical crystal structure of layered cathode materials, which belongs to the space group of $R\bar{3}m$ (JCPDS 82-1495) [41,42]. Unlike PC-NCM particles, the (003) and (104) peak shifts were clearly observed in SC-NCM particles because of the decreased lattice parameters, as shown in Table S1. The reduction in the lattice parameters was attributed to the Li deficiency in the NCM structure with a Li/TM ratio of less than 1, which could be induced by the relatively high synthesis temperature of SC-NCM particles. The Li loss did not occur during wet milling but occurred during the post-heat treatment of SC-NCM particles. As shown in Fig. 3b, the microstructures

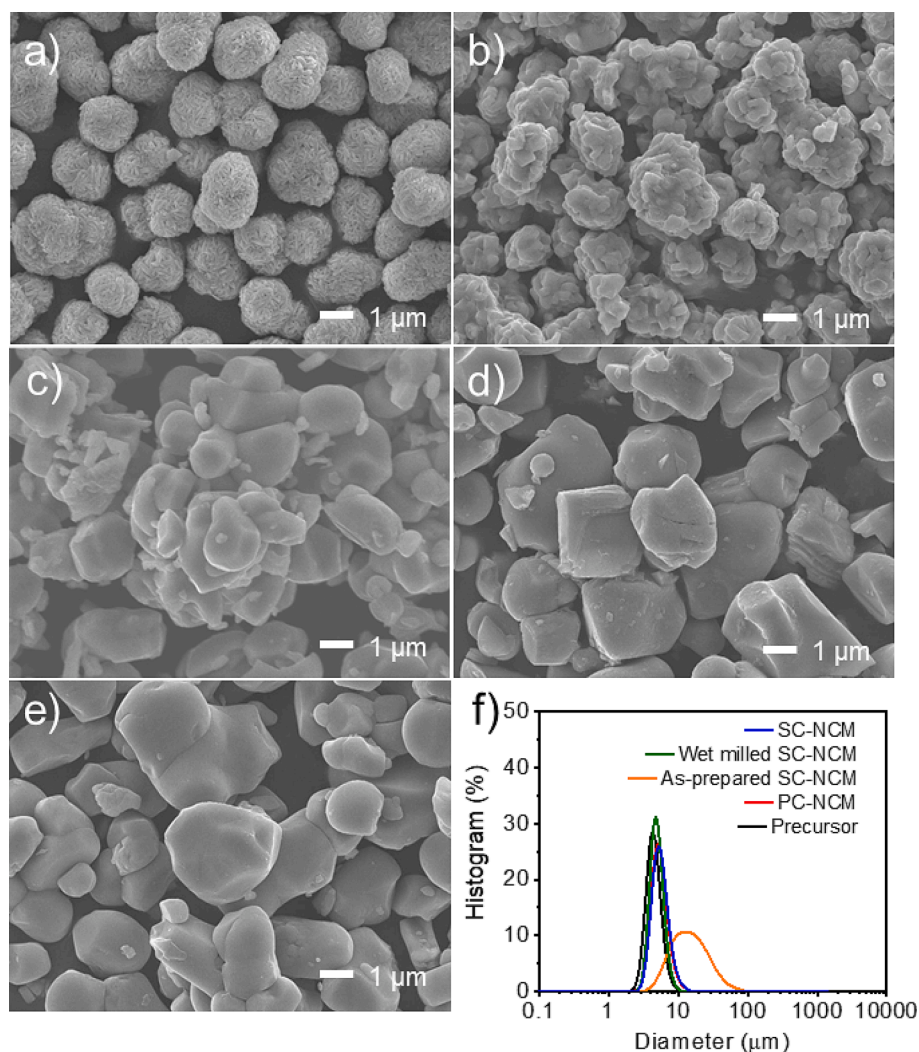


Fig. 2. FESEM images of precursor and NCMs with different synthesis conditions: (a) precursor (b) 800 °C (PC-NCM), (c) 900 °C (as-prepared SC-NCM), (d) wet-milled SC-NCM, (e) SC-NCM after a post-heat treatment, and (f) comparison on particle size distributions of the samples.

of PC-NCM and SC-NCM particles were observed using TEM. As expected, PC-NCM particles contained a large number of structural defects (i.e. internal voids and grain boundaries), whereas there was no evidence of grain boundaries or internal voids in SC-NCM particles. The microstructural difference between PC-NCM and SC-NCM particles was mainly attributed to the different process variables during synthesis. High-resolution TEM images and fast Fourier transform (FFT) patterns confirmed the fine crystal structures of PC-NCM and SC-NCM particles.

The mechanical strengths of PC-NCM and SC-NCM particles were compared by measuring particle hardness using a micro-indenter. The stress-strain curves of PC-NCM and SC-NCM particles (Fig. 3c) showed that the hardness of SC-NCM particles (972.7 MPa) was considerably higher than that of PC-NCM particles (~113.3 MPa). This is mainly attributed to the large crystallite size and high density of SC-NCM particles owing to their distinctive morphology and microstructure. As a high stack pressure is typically applied to maintain stable interfacial contact in the electrodes of ASSBs, the particle hardness of cathode materials is crucial for ensuring long-term cycling stability.

The electrochemical performance of PC-NCM and SC-NCM cathodes was examined in the LIB assembled with a conventional liquid electrolyte. The galvanostatic voltage profiles from 3.0 V to 4.3 V vs. Li/Li⁺ (Fig. S6) showed that both cathodes exhibited similar electrochemical behaviors toward Li⁺ insertion and extraction at current densities of 0.1–5.0 C (1 C = 200 mA g⁻¹). As expected, the reversible capacities of

the PC-NCM cathode were relatively higher than those of the SC-NCM cathode, regardless of the applied current density. However, the capacity retention of the SC-NCM cathode after 100 cycles was slightly higher than that of the PC-NCM cathode at a current density of 0.5 C.

To clarify the reason for the different electrochemical behaviors of the PC-NCM and SC-NCM cathodes, the cut-off voltage was increased to 4.5 V vs. Li/Li⁺. The reversible capacities of the PC-NCM cathode were higher than those of the SC-NCM cathode at various current densities, as shown in Fig. 4a and 4b.

To examine the kinetics of Li intercalation into PC-NCM and SC-NCM, the dependence of cathodic polarization (ΔV) on the state of charge (SOC) was determined using the galvanostatic intermittent titration technique (GITT). Here, ΔV represents the voltage decay due to internal resistances during Li⁺ intercalation, which arises from the ohmic resistance as well as from interfacial charge transfer and chemical diffusion processes. The results demonstrated that the PC-NCM cathode exhibited a smaller ΔV compared to the SC-NCM cathode over the whole range of SOC (Fig. 4c). Additionally, electrochemical impedance spectroscopy (EIS) was utilized to assess the interfacial resistance, as shown in Fig. S7. The charge transfer resistance (R_{ct}), represented by the second semicircle observed at middle-to-low frequencies, was found to be lower for PC-NCM compared to SC-NCM. The observed differences in ΔV and R_{ct} between PC-NCM and SC-NCM can be reasonably attributed to their distinct morphological and

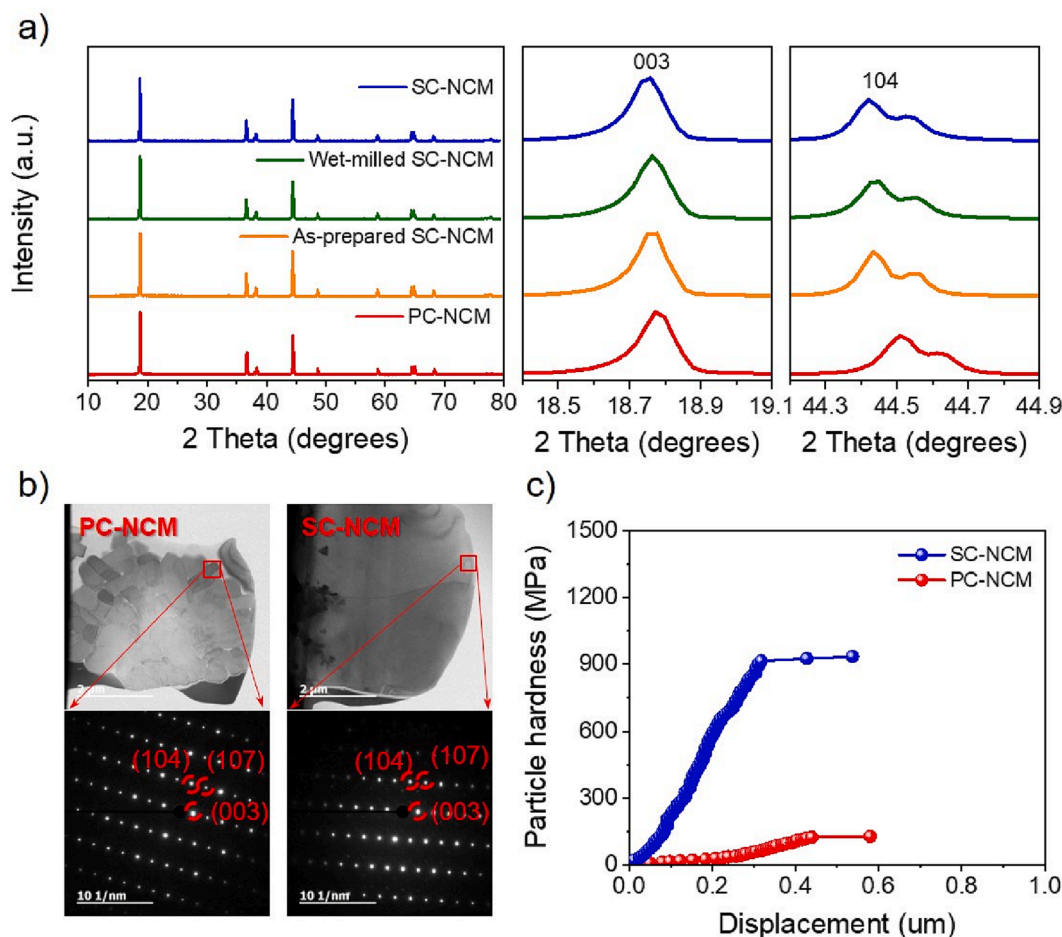


Fig. 3. Structural characterizations of NCMs: comparisons of (a) XRD patterns, (b) TEM images with corresponding FFT patterns, (c) particle hardness of NCMs.

microstructural characteristics. In the case of PC-NCM, the liquid electrolyte readily penetrates between the primary particles, thereby reducing R_{ct} by providing a larger interface for Li^+ insertion and extraction. Moreover, the penetration of the liquid electrolyte allows for shorter diffusion lengths within the bulk of PC-NCM. As a result, the PC-NCM cathode exhibits superior rate capability and lower overpotential when used in LIB with a conventional liquid electrolyte [43].

However, the hardness of PC-NCM particles decreased considerably after soaking in the liquid electrolyte. In contrast, SC-NCM particles maintained their high particle hardness even after soaking (Fig. S8). We expected that this will be beneficial for improving the cycling stability of the SC-NCM cathode because the formation of cracks and side reactions can be mitigated owing to its excellent mechanical strength [17,44]. In practice, the capacity retention of the SC-NCM cathode (75 %) was relatively higher than that of the PC-NCM cathode (62.9 %) after 100 cycles at a current density of 0.5 C (Fig. 4d). This improvement is mainly associated with the enhanced hardness of SC-NCM particles. It is well known that cracks are generated under mechanical stress, which can be caused by 1) a change in volume during cycling and 2) pressure during electrode fabrication.

The cross-sectional SEM images of cycled PC-NCM and SC-NCM cathodes were compared in Fig. 5a and 5b, respectively. The formation of microcracks was clearly observed in PC-NCM particles. In contrast, microcrack formation was successfully mitigated in SC-NCM particles after cycling. As the liquid electrolyte can easily penetrate microcracks, it leads to undesirable side reactions such as the corrosion of grain boundaries and metal dissolution. This resulted in performance fading. As confirmed by the differential voltage (dQ/dV) curves, the SC-NCM cathode exhibited a lower overpotential than the PC-NCM cathode

after cycling. This indicated that the structural degradation of the SC-NCM cathode was suppressed by its high mechanical strength (Fig. 5c and 5d). Thus, the excellent hardness of SC-NCM particles should be considered as a predominant factor that directly affects the cycling performance.

The feasibility of the SC-NCM cathode was examined in an ASSB assembled with a composite cathode composed of SC-NCM particles, a conducting agent (Super P), and an LPSCI solid electrolyte (60:5:35).

During the optimization process, the rate capabilities of as-prepared SC-NCM, wet-milled SC-NCM, and post-heat-treated SC-NCM cathodes were evaluated within a voltage range of 1.9–3.7 V vs. In-Li/Li^+ at room temperature (Fig. S9). Upon comparison, we found that both as-prepared SC-NCM and wet-milled SC-NCM cathodes exhibited similar electrochemical behaviors under various current densities ranging from 0.1 to 5.0 C. This implies that there was a balanced impact of particle size and surface damage on the electrochemical performance of these cathode materials, leading to comparable performance. More importantly, regardless of the applied current density, the post-heat-treated SC-NCM cathode showed much improved capacities, which indicates that the optimization process positively impacted the electrochemical performance of SC-NCM.

Fig. 6a and b showed the galvanostatic charge and discharge profiles of the ASSBs configured with the PC-NCM and SC-NCM cathodes, respectively. The capacity and initial coulombic efficiency of the SC-NCM cathode (187.0 mA g^{-1} and 79.9 %) were relatively higher than those of the PC-NCM cathode (185.0 mA g^{-1} and 78.3 %) in the first cycle recorded at a current density of 0.05C. The reversible capacities of the SC-NCM cathode were also higher than those of the PC-NCM cathode at current densities of 0.05, 0.1, 0.2, 0.5, 1, 2, and 5 C. This is opposite to

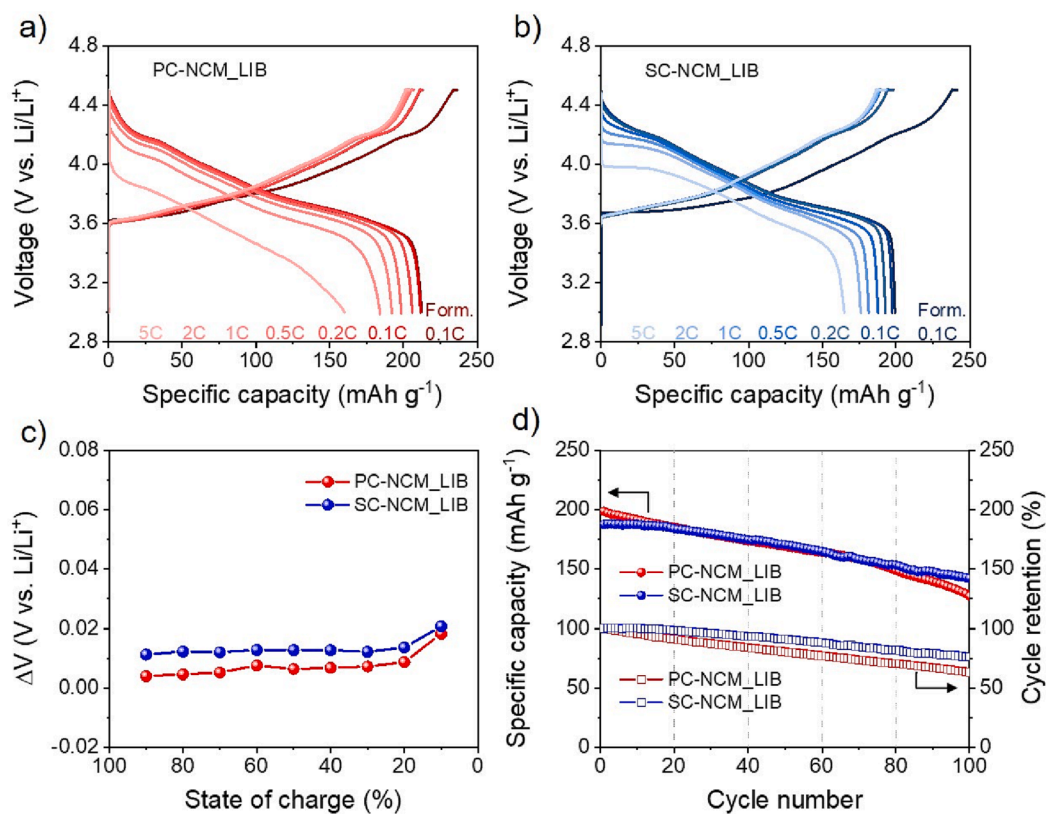


Fig. 4. Electrochemical performance of PC-NCM and SC-NCM in LIBs: voltage profiles of (a) PC-NCM and (b) SC-NCM. (c) SOC dependence of cathodic polarization (ΔV). (d) Cycling performance (0.5 C).

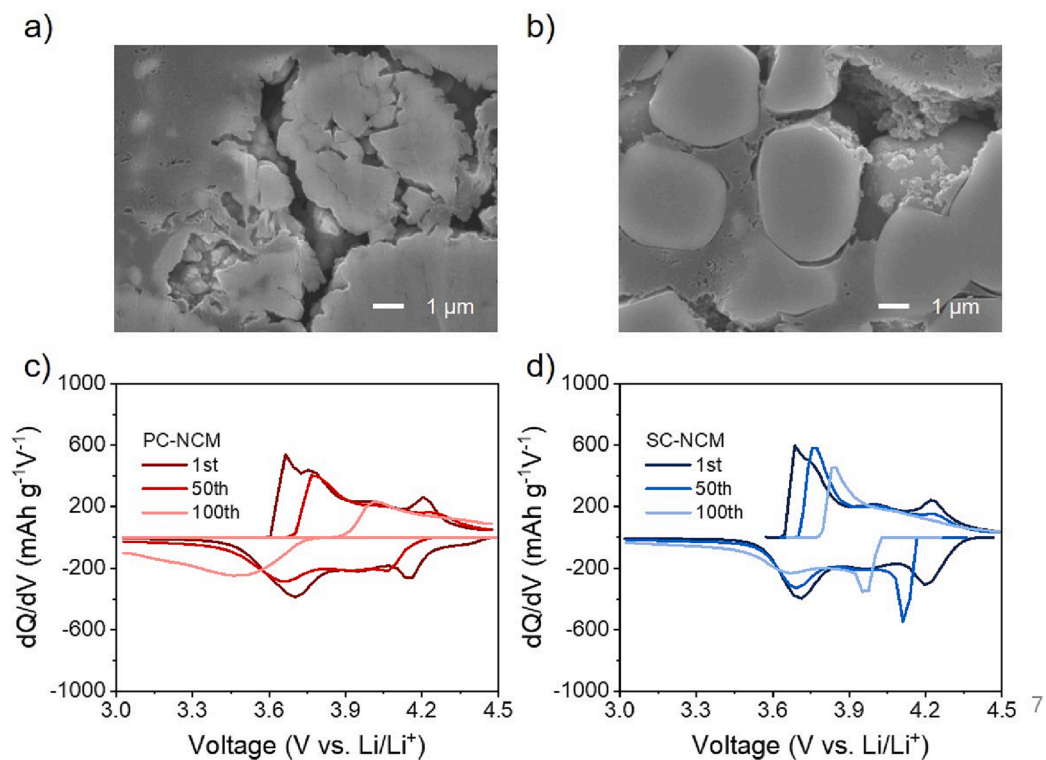


Fig. 5. Cross-sectional FESEM images of cathodes after 100 cycles in LIBs: (a) PC-NCM and (b) SC-NCM. (c) dQ/dV graphs for 1st, 50th, and 100th cycles of PC-NCM and (d) SC-NCM.

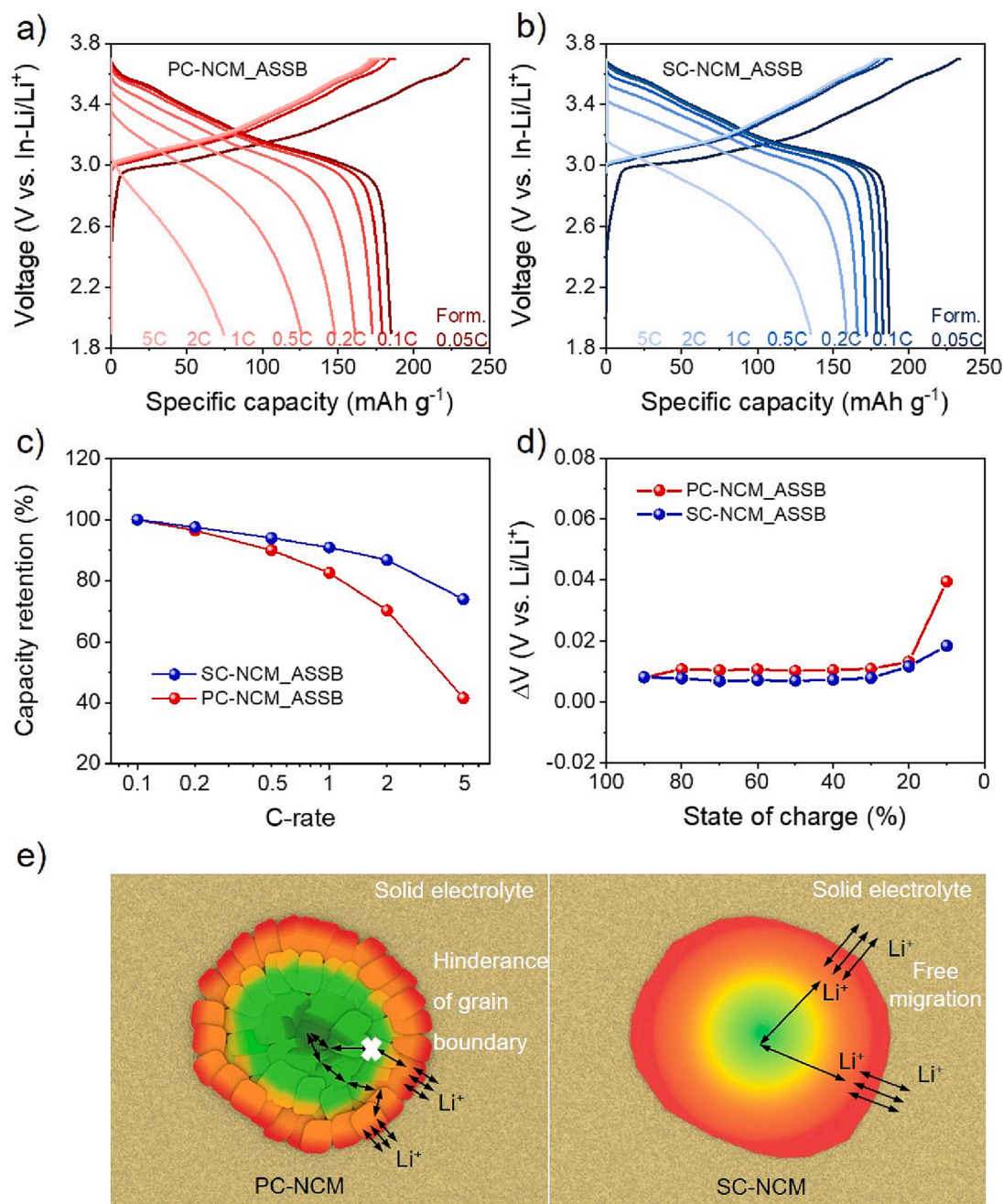


Fig. 6. Electrochemical performance of PC-NCM and SC-NCM in ASSBs: voltage profiles of (a) PC-NCM and (b) SC-NCM. (c) rate capabilities at various C-rates. (d) SOC dependence of cathodic polarization (ΔV). (e) schematics of the Li^+ transport pathway from surface to bulk for PC-NCM and SC-NCM.

the trend observed for a conventional LIB with a liquid electrolyte. Even at a high current density (5 C), we noted that the capacity retention of the SC-NCM cathode was 74.0 %, while a considerably lower value was retained in the PC-NCM cathode (41.6 %) (Fig. 6c). As shown in Fig. 6d, the trend of the ΔV dependence on the SOC of the ASSB was opposite to that observed for the LIB (Fig. 4c).

The abovementioned results showed that the rate capabilities of the ASSB can be improved by employing the SC-NCM cathode materials. Fig. 6e shows the different microstructures of PC-NCM and SC-NCM particles with different reaction mechanisms in the ASSB. In a conventional LIB, the large active interface and short diffusion lengths in primary particles of PC-NCM are beneficial for reducing the interfacial and diffusion resistances, thereby increasing the reversible capacities. This is because each primary particle can offer topotactic Li^+ insertion and extraction for PC-NCM particles, thereby enhancing the kinetics during

cycling. In contrast, charge-transfer reactions occur only along the surface of SC-NCM particles with a small active interface, which mainly limits the reversible capacity during cycling. Nevertheless, the benefits of PC-NCM particles are not applicable in an ASSB with a solid electrolyte because the solid electrolyte cannot penetrate the cathode particles. In this respect, the interfacial contact between the cathode particles and solid electrolyte is only formed on the outermost surface of the cathode particles. Therefore, the Li^+ transport pathways from the surface of particle to the bulk govern the overall electrochemical reaction in the ASSB.

In PC-NCM particles, Li^+ diffusion can be restricted by grain boundaries and internal voids between the primary particles. In contrast, defect-free SC-NCM particles allow fast Li^+ diffusion during cycling. Furthermore, the undesirable particle cracking and contact loss of SC-NCM particles can be suppressed by their extremely high particle

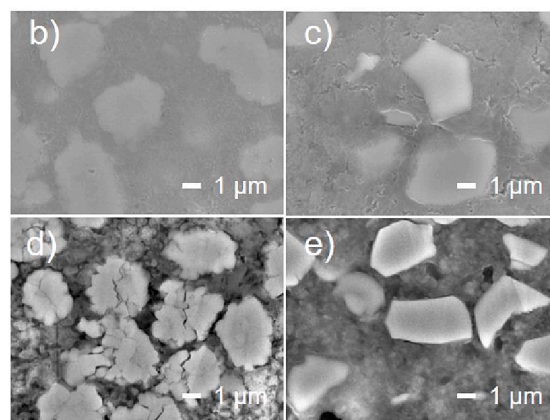
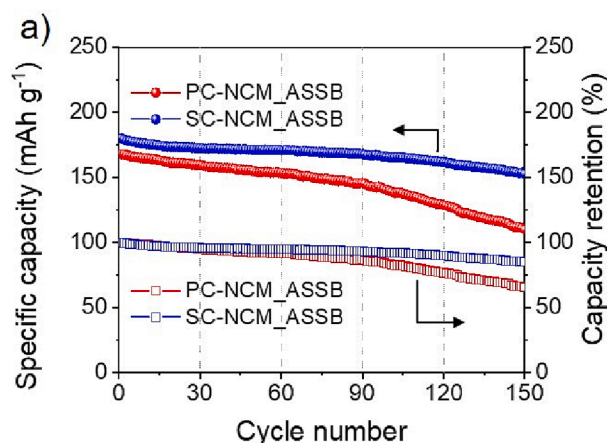


Fig. 7. Cycling performance of ASSBs and FESEM images of cathodes: (a) ASSBs with PC-NCM and SC-NCM cathodes at cut-off voltage 3.9 V vs. In-Li/Li⁺. (b) PC-NCM and (c) SC-NCM before cycling. (d) PC-NCM and (e) SC-NCM after 150 cycles.

hardness. Therefore, we note that effective Li⁺ transport pathways can be developed in the SC-NCM cathode. As compared in Fig. 7, the capacity retention and discharge capacity of the SC-NCM cathode (84.9%) were higher than those of the PC-NCM cathode (65.6%) after 150 cycles at 0.5 C with a cut-off voltage of 3.9 V vs. In-Li/Li⁺ at room temperature. Cross-sectional SEM images indicate the formation of microcracks in PC-NCM particles and the effective suppression of microcracks in SC-NCM particles after 150 cycles.

SC-NCM particles with higher structural integrity and mechanical strength are more suitable for ensuring the reversibility and long-term cycling performance of an ASSB configured with a solid electrolyte. In practice, the performance degradation of ASSBs is induced by the contact loss of cathode particles with a solid electrolyte owing to the cracking and insolation of cathode particles during cycling. Therefore, we emphasize that the structural engineering of cathode particles is required to make them more suitable for boosting Li⁺ transport and building stable interfaces with solid electrolytes in the development of high-performance ASSBs.

4. Conclusion

The distinctive structural characteristics of SC-NCM cathode materials with excellent mechanical strength are advantageous for enhancing the reversibility and long-term cycling stability of ASSBs. In practice, the interfacial resistance at the solid-solid interfaces between LPSCI solid electrolyte and cathode materials can be effectively reduced by the structural modification of Ni-rich cathode materials in the form of single-crystalline. Robust Li⁺ pathways and stable interfaces with solid electrolytes can be established in electrodes using single-crystalline cathode materials. Moreover, the excellent mechanical strength of single-crystalline cathode materials is mainly responsible for the improvement in the long-term cycling stability of ASSBs. This study provides practical insights into the development of highly reliable cathode materials for commercial ASSB applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2023.144381>.

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